

YINING, China

WMO: 514310

Lat: 43.9554N

Lon: 81.3363E

Elev: 2178

StdP: 13.58

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.9	(c) -0.7	(d) -10.4	(e) 3.4	(f) -5.3	(g) -5.2	(h) 4.5	(i) 0.1	(j) 12.3	(k) 23.9	(l) 9.7	(m) 25.8	(n) 1.6	(o) 70	(p) 0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 22.5	(c) 93.3	(d) 66.3	(e) 90.7	(f) 65.5	(g) 88.3	(h) 64.6	(i) 69.3	(j) 86.4	(k) 68.0	(l) 85.3	(m) 66.7	(n) 83.5	(o) 4.2	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.0	96.9	75.8	62.3	91.1	74.4	60.8	86.4	73.5	34.9	86.4	33.7	85.6	32.6	83.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.9	(b) 11.1	(c) 9.2	(e) -9.9	(f) 97.8	(g) 6.3	(h) 1.7	(i) -14.4	(j) 99.1	(k) -18.1	(l) 100.0	(m) -21.6	(n) 101.0	(o) -26.2	(p) 102.2		
(4) 13.9	(5) 11.1	(6) 9.2	(7) -9.9	(8) 97.8	(9) 6.3	(10) 1.7	(11) -14.4	(12) 99.1	(13) -18.1	(14) 100.0	(15) -21.6	(16) 101.0	(17) -26.2	(18) 102.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	18.8	25.5	42.3	56.0	64.9	72.2	75.2	73.1	64.4	51.9	38.5	24.4		
	DBStd	20.83	9.11	8.80	9.70	7.91	6.29	4.98	4.49	5.00	6.35	7.40	8.29	8.55		
	HDD50	3177	966	686	275	36	1	0	0	0	1	65	353	794		
	HDD65	6165	1431	1106	705	279	80	7	1	5	87	409	796	1259		
	CDD50	3441	0	0	35	217	461	667	782	716	433	123	7	0		
	CDD65	954	0	0	0	10	75	224	318	256	69	2	0	0		
	CDH74	10455	0	0	9	229	888	2198	3289	2847	935	60	0	0		
	CDH80	4131	0	0	0	49	248	831	1462	1246	289	6	0	0		
(14) Wind	WSAvg	3.8	2.6	3.1	4.4	5.1	4.6	4.5	4.2	3.9	3.5	3.2	3.1	2.8		
(15) Precipitation	PrecAvg	10.7	0.7	0.7	0.8	1.1	1.2	1.1	1.0	0.6	0.6	1.0	1.2	0.9		
	PrecMax	18.5	3.7	2.6	3.4	3.9	2.6	2.8	3.1	1.8	1.9	3.4	4.5	2.7		
	PrecMin	5.4	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.1	0.1	0.0		
	PrecStd	3.2	0.6	0.5	0.6	0.7	0.6	0.7	0.7	0.4	0.4	0.7	0.9	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.3	50.1	74.0	85.3	89.7	93.3	97.5	96.0	90.4	80.0	66.2	48.1		
		MCWB	35.0	40.8	51.6	58.0	62.0	65.1	68.6	66.8	64.5	57.8	49.3	38.4		
	2%	DB	37.3	44.4	67.6	80.1	85.9	90.3	94.0	92.6	86.6	74.8	60.2	42.3		
		MCWB	32.0	36.8	49.9	55.6	61.0	64.7	67.2	66.1	62.4	55.6	46.8	35.1		
	5%	DB	34.4	40.5	63.1	75.9	82.6	87.8	91.1	90.1	83.4	70.2	54.6	38.8		
		MCWB	30.6	34.5	48.2	54.4	60.0	64.5	66.1	65.5	61.1	53.5	44.8	33.3		
	10%	DB	32.0	37.3	58.3	71.6	79.2	85.2	88.0	87.3	79.8	65.8	50.2	35.3		
		MCWB	29.1	32.5	46.1	52.9	58.7	63.5	65.3	64.7	59.8	51.6	42.6	31.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.0	41.3	53.7	61.1	66.3	69.9	72.2	70.7	66.5	59.0	51.6	39.6		
		MCDB	40.3	48.6	69.2	78.6	83.2	85.8	89.4	87.2	86.2	76.9	60.8	45.7		
	2%	WB	33.2	37.6	51.1	57.9	63.6	67.9	69.9	68.9	64.0	56.8	48.2	36.4		
		MCDB	36.2	43.0	65.3	75.1	80.0	83.9	87.5	86.0	82.6	72.0	57.7	41.4		
	5%	WB	31.3	35.2	48.9	55.8	61.9	66.5	68.4	67.5	62.3	54.9	45.6	33.7		
		MCDB	33.8	39.5	61.6	71.1	77.9	82.6	85.8	85.2	80.3	67.4	53.7	37.7		
	10%	WB	29.1	33.2	46.7	54.3	60.3	65.2	67.1	66.0	60.6	52.9	43.2	31.8		
		MCDB	31.8	36.6	56.6	68.0	75.2	80.7	83.7	83.3	77.1	63.4	49.7	34.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	16.7	18.9	22.1	22.3	21.5	22.5	24.6	25.9	24.2	17.8	16.7		
		MCDBR	19.8	19.5	26.4	30.6	27.7	25.9	27.5	29.4	31.6	31.2	26.7	21.0		
		MCWBR	14.7	12.8	13.1	12.6	10.4	9.2	9.4	11.1	13.0	13.8	14.3	14.3		
		MCDBR	15.6	16.4	24.2	25.0	23.3	22.2	24.0	26.1	29.4	27.0	24.0	17.7		
	5% WB	MCWBR	11.6	10.9	12.3	10.8	9.4	8.6	9.4	10.7	12.8	12.1	13.2	12.1		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.371	0.432	0.472	0.443	0.439	0.426	0.404	0.391	0.384	0.361	0.321		
	taud		2.174	2.074	1.973	1.922	2.068	2.144	2.208	2.256	2.230	2.153	2.139	2.178		
	Ebn at Noon		245	251	250	251	264	265	267	269	262	244	226	232		
	Edh at Noon		30	40	50	57	51	48	44	41	38	36	31	27		
(44) All-Sky Solar Radiation	RadAvg		678	978	1302	1634	1992	2047	2042	1892	1547	1023	638	539		
	RadStd		30	48	87	127	115	81	115	89	87	94	52	35		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	+0.99	+3.92	N/A	+0.85	-0.97	-1.40	-256	-297	+211	+115
Regional (2 neighbors)	N/A	N/A	N/A	N/A	-0.76	N/A	N/A	N/A	+169	N/A

Nomenclature: See separate page